Digital images are an essential part of computer science and engineering. They are used in web development, photography, medical imaging, computer vision, multimedia applications, and scientific visualization.

An image file format defines how image data is encoded, compressed, and stored in a computer system. Choosing the right format is important because:

- Some applications require lossless storage (e.g., medical X-rays, where every pixel matters).
- Others prefer small size with acceptable quality (e.g., social media photos).
- Some applications need scalable graphics (e.g., logos on websites).

Thus, understanding the properties of common image formats is critical for engineers.

# Image Representation in Computers

# 1. Raster Images

A raster image is a grid (matrix) of tiny dots called pixels (picture elements). Each pixel contains color information, usually stored in RGB (Red, Green, Blue) or CMYK (Cyan, Magenta, Yellow, Black for printing).

**Characteristics**

- Resolution-dependent: Quality depends on the number of pixels (resolution). If scaled up, the image becomes pixelated (blurry, blocky).
- File Size: Can be large because each pixel stores data.
- Color Depth: Usually 8-bit (256 colors), 24-bit (16.7 million colors), or 32-bit (with transparency channel).

**Examples of Raster Formats:** BMP, JPEG, PNG, GIF, TIFF.

**Advantages**

- Can represent very detailed and realistic images.
- Good for photographs and complex textures.
- Supported universally across devices and platforms.

**Disadvantages**

- Large file sizes for high-resolution images.
- Scaling reduces quality (loss of sharpness).
- Editing (resizing, cropping) can degrade image quality.

**Applications**

- Digital photography.
- Web images and graphics.
- Printing photographs.
- Medical imaging (X-rays, CT scans).

# 2. Vector Images

A vector image is made of geometric primitives — lines, curves, polygons — defined using mathematical equations instead of pixels. A circle in a vector image is stored as an equation (center + radius), not as thousands of colored pixels.

**Characteristics**

- **Resolution-independent**: Can be scaled up or down infinitely without losing quality.
- **Small file size**: Stores shapes and properties, not individual pixels.
- **Editable**: Each shape, line, or curve can be easily modified.

**Examples of Vector Formats:** SVG (Scalable Vector Graphics), EPS, PDF (for diagrams), AI (Adobe Illustrator).

**Advantages**

- Infinitely scalable without pixelation.
- Small file size for line art and diagrams.
- Easy to edit (change colors, shapes, size).

**Disadvantages**

- Cannot represent highly detailed images like photographs.
- Requires special software for editing.
- Rendering may be slower for very complex vector graphics.

**Applications**

- Logos and icons.
- Engineering diagrams and CAD drawings.
- Maps, charts, and infographics.
- Fonts (letters are vector shapes).

# Comparing Raster vs Vector: Key Differences.

**Resolution:**

- Raster images are resolution-dependent; enlarging them results in pixelation.
- Vector images are resolution-independent; they stay sharp at any size.

**Detail and complexity:**

- Raster excels at complex images like photographs with shading and gradients.
- Vector is ideal for simple, clean graphics like logos and icons.

**File size:**

- Raster files can be large, especially at high resolutions.
- Vector files are generally smaller and more efficient.

**Software compatibility:**

- Raster: Best edited in Photoshop.
- Vector: Best created and edited in Illustrator.

# Common Raster Image Formats

**A. BMP (Bitmap Image File)**

**History:** Developed by Microsoft for Windows operating systems.

**Technical Details:**

- Stores pixel data uncompressed.
- Each pixel directly mapped in binary (e.g., 24-bit = 3 bytes per pixel).
- Very simple structure, but very large files.

**Advantages:**

- Exact pixel-by-pixel accuracy.
- Easy for programming and image manipulation.

**Disadvantages:**

- Extremely large file sizes.
- Not suitable for sharing or web use.

**Applications:**

- Image processing labs.
- When raw, unaltered image data is needed.

**B. JPEG (Joint Photographic Experts Group)**

**History:** Introduced in 1992, one of the most widely used formats.

**Technical Details:**

- Uses lossy compression (Discrete Cosine Transform).
- Removes imperceptible visual details to reduce size.
- Compression level adjustable → smaller size but lower quality.

**Advantages:**

- Very small size (compared to BMP).
- Ideal for photos and web use.

**Disadvantages:**

- Each save reduces quality (lossy).
- Not suitable for images needing fine detail (e.g., medical scans, text).

**Applications:**

- Digital photography.
- Online sharing.
- Social media platforms.

**C. PNG (Portable Network Graphics)**

**History:** Created in 1996 as an improved replacement for GIF.

**Technical Details:**

- Lossless compression (Deflate algorithm).
- Supports transparency with alpha channels (0–255 levels).
- Better compression than BMP but not lossy like JPEG.

Advantages:

- High-quality images with no loss.

- Excellent for graphics, line art, and transparency.

**Disadvantages:**

- Larger size compared to JPEG for photos.

**Applications:**

- Logos, icons, screenshots.
- Web images needing transparent backgrounds.

**D. GIF (Graphics Interchange Format)**

**History:** Created in 1987 by CompuServe.

**Technical Details:**

- 8-bit color (256 colors only).
- Uses LZW lossless compression.
- Supports simple animations.
- Transparency supported, but only one-bit (fully transparent or not).

**Advantages:**

- Very small size for low-color images.
- Supports animation.

**Disadvantages:**

- Limited color palette (unsuitable for photos).

**Applications:**

- Animated memes, web banners.
- Simple icons.

**E. TIFF (Tagged Image File Format)**

**History:** Developed by Aldus (now Adobe) in the 1980s.

**Technical Details:**

- Supports both lossy and lossless compression.
- Can store multiple images (multi-page format).

- Rich metadata support.

**Advantages:**

- Extremely high quality.
- Widely used in professional industries.

**Disadvantages:**

- Very large file sizes.

**Applications:**

- Medical imaging (MRI, CT scans).
- Publishing and printing.
- Archiving documents.

# Vector Image Format

**SVG (Scalable Vector Graphics)**

**History:** Developed by W3C in 1999.

**Technical Details:**

- Based on XML markup language.
- Describes images using mathematical shapes, not pixels.
- Infinite scalability (no quality loss).

**Advantages:**

- Very small file size for line art.
- Editable in text editors and scalable.

**Disadvantages:**

- Not suitable for photos.

**Applications:**

- Logos, diagrams, charts, maps.
- Numerical Example:
- A company logo in SVG: 10 KB, vs PNG version: 200 KB.

# Image Color Models

A color model is a mathematical or conceptual framework for representing colors in digital or physical mediums. Different models are optimized for different devices, processes, or use cases—such as screens (additive light) vs. printers (subtractive ink), or intensity-based processing (grayscale) vs. human-perception adjustments (HSV).

## RGB Color Model

In the RGB model, each color appears as a combination of red, green, and blue. This model is called additive, and the colors are called primary colors. The primary colors can be added to produce the secondary colors of light (see Figure 1) - magenta (red plus blue), cyan (green plus blue), and yellow (red plus green). The combination of red, green, and blue at full intensities makes white.

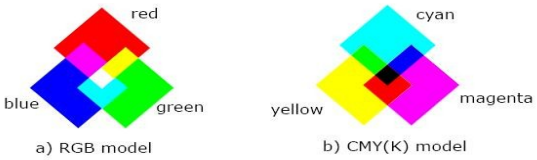



Figure 1: Colors for RGB and CMY(K) Models

The color subspace of interest is a cube shown in Figure 2 (RGB values are normalized to 0..1), in which RGB values are at three corners; cyan, magenta, and yellow are the three other corners, black is at their origin; and white is at the corner farthest from the origin.The gray scale extends from black to white along the diagonal joining these two points. The colors are the points on or inside the cube, defined by vectors extending from the origin.

Thus, images in the RGB color model consist of three independent image planes, one for each primary color. As a rule, the Intel IPP color conversion functions operate with non-linear gamma-corrected images R'G'B'. The importance of the RGB color model is that it relates very closely to the way that the human eye perceives color. RGB is a basic color model for computer graphics because color displays use red, green, and blue to create the desired color. Therefore, the choice of the RGB color space simplifies the architecture and design of the system. Besides, a system that is designed using the RGB color space can take advantage of a large number of existing software routines, because this color space has been around for a number of years.

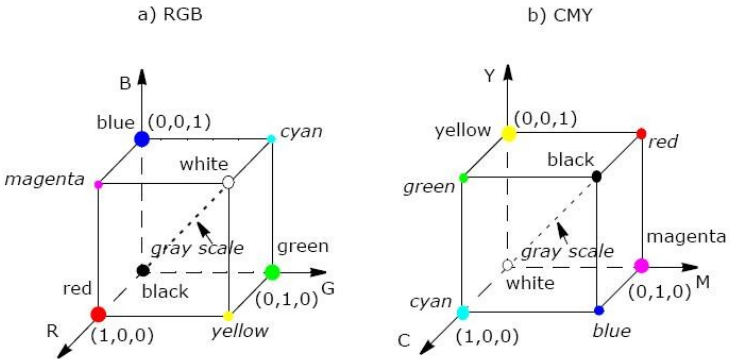



Figure 2: RGB and CMY Color Models

However, RGB is not very efficient when dealing with real-world images. To generate any color within the RGB color cube, all three RGB components need to be of equal pixel depth and display resolution. Also, any modification of the image requires modification of all three planes.

**CMYK Color Model**

The CMYK color model is a subset of the RGB model and is primarily used in color print production. CMYK is an acronym for cyan, magenta, and yellow along with black (noted as K). The CMYK color space is subtractive, meaning that cyan, magenta yellow, and black pigments or inks are applied to a white surface to subtract some color from white surface to create the final color. For example (see Figure 1"), cyan is white minus red, magenta is white minus green, and yellow is white minus blue. Subtracting all colors by combining the CMY at full saturation should, in theory, render black. However, impurities in the existing CMY inks make full and equal saturation impossible, and some RGB light does filter through, rendering a muddy brown color. Therefore, the black ink is added to CMY. The CMY cube is shown in Figure 2, in which CMY values are at three corners; red, green, and blue are the three other corners, white is at the origin; and black is at the corner farthest from the origin.

# What is Image Processing?

Image processing is a method used to perform operations on an image to enhance it or to extract useful information from it. It involves various techniques and algorithms that process images in a digital format. This can include a range of tasks such as improving the visual quality of images, detecting patterns, segmenting objects, and transforming images into different formats. Image processing can be used for both photos and video frames. The process usually involves steps such as inputting the image, processing the image through various algorithms, and then outputting the results in a format that is usable or can be further analyzed.

# Types of Image Processing

## 1. Analog Image Processing

Analog image processing refers to techniques used to process images in their analog form, such as photographs, printed pictures, or images captured on film. This type of processing involves modifying images through physical or chemical means. Before the advent of digital technology, all image processing was done using analog methods. These methods are generally less flexible and more time-consuming compared to digital techniques, but they have historical significance and specific applications.

## 2. Digital Image Processing

Digital image processing involves the use of computer algorithms to perform operations on digital images. Unlike analog processing, digital techniques offer more flexibility, precision, and automation. Digital images are composed of pixels, and processing these images involves manipulating pixel values to achieve the desired effect. The use of digital processing is widespread due to its efficiency and the vast array of tools and techniques available.

# Image Processing Techniques

## 1. Image Enhancement

### 1. Contrast Adjustment

Contrast adjustment is a technique used to improve the visibility of features in an image by enhancing the difference between the light and dark areas. This can be achieved through methods like contrast stretching, which adjusts the intensity values of pixels to span the full range of the histogram.

### 2. Histogram Equalization

Histogram equalization is a method used to enhance the contrast of an image by transforming its intensity values so that the histogram of the output image is evenly distributed. This technique improves the global contrast and is particularly useful in images with backgrounds and foregrounds that are both bright or both dark.

### 3. Noise Reduction

Noise reduction techniques are used to remove unwanted random variations in brightness or color, known as noise, from an image. Common methods include median filtering, Gaussian smoothing, and bilateral filtering, each of which aims to smooth the image while preserving important details.

## 2. Image Restoration

### 1. Deblurring

Deblurring techniques are used to restore sharpness to an image that has been blurred due to factors like camera shake or motion. Methods such as inverse filtering and Wiener filtering are commonly employed to reconstruct the original image.

### 2. Inpainting

Inpainting involves reconstructing lost or deteriorated parts of an image. This technique is often used for restoring old photographs, removing objects, or filling in missing data. Algorithms for inpainting include patch-based methods and partial differential equations (PDE) based methods.

### 3. Denoising

Denoising is the process of removing noise from an image while preserving its details. Techniques such as wavelet thresholding and non-local means filtering are used to achieve this, ensuring that the image quality is improved without losing significant features.

## 3. Image Segmentation

### 1. Thresholding

Thresholding is a simple technique for segmenting an image by converting it into a binary image. This is done by selecting a threshold value, and all pixels with intensity values above the threshold are turned white, while those below are turned black.

### 2. Edge Detection

Edge detection involves identifying the boundaries within an image. Techniques like the Sobel, Canny, and Prewitt operators are used to detect edges by finding areas of high intensity gradient.

### 3. Region-Based Segmentation

Region-based segmentation divides an image into regions based on predefined criteria. This can include methods like region growing, where adjacent pixels are grouped based on similar properties, and watershed segmentation, which treats the image like a topographic map.

## 4. Image Compression

### 1. Lossy Compression

Lossy compression reduces the size of an image file by permanently eliminating certain information, especially redundant data. Techniques like JPEG compression are used to significantly reduce file size at the cost of some loss in quality.

### 2. Lossless Compression

Lossless compression reduces the image file size without any loss of quality. Methods such as PNG compression ensure that all original data can be perfectly reconstructed from the compressed file.

## 5. Image Synthesis

### 1. Texture Synthesis

Texture synthesis generates large textures from small sample images, ensuring that the generated texture looks natural and continuous. This technique is widely used in computer graphics and game design.

### 2. Image Generation

Image generation involves creating new images from scratch or based on existing images using techniques such as generative adversarial networks (GANs). This can be used in applications like creating realistic human faces or artistic images.

## 5. Feature Extraction

### 1. Shape and Texture Analysis

Shape and texture analysis techniques are used to identify and quantify the shapes and textures within an image. Methods like edge detection, contour analysis, and texture filters help in understanding the geometric and surface properties of objects in the image.

### 2. Color Detection

Color detection involves identifying and segmenting objects based on their color properties. Techniques such as color thresholding and color histograms are used to analyze the color distribution and extract relevant features.

### 3. Pattern Recognition

Pattern recognition is the process of classifying input data into objects or classes based on key features. Techniques such as neural networks, support vector machines, and template matching are used to recognize patterns and make classifications.

## 6. Morphological Processing

### 1. Dilation and Erosion

Dilation and erosion are basic morphological operations used to process binary images. Dilation adds pixels to the boundaries of objects, making them larger, while erosion removes pixels from the boundaries, making objects smaller.

### 2. Opening and Closing

Opening and closing are compound operations used to remove noise and smooth images. Opening involves erosion followed by dilation, which removes small objects and smooths contours. Closing involves dilation followed by erosion, which fills small holes and gaps.

### 3. Morphological Filters

Morphological filters are used to process images based on their shapes. These filters, including hit-or-miss transform and morphological gradient, are used to extract relevant structures and enhance image features.

# Applications of Image Processing

## 1. Medical Imaging

- **MRI and CT Scans:** Enhancing the clarity of MRI and CT scans for better diagnosis and treatment planning.
- **X-Ray Imaging:** Improving the quality and detail of X-ray images to detect fractures, tumors, and other anomalies.
- **Ultrasound Imaging:** Enhancing ultrasound images for more accurate visualization of internal organs and fetal development.

## 2. Remote Sensing

- **Satellite Imaging:** Analyzing satellite images for applications like land use mapping and resource monitoring.
- **Aerial Photography:** Using drones and aircraft to capture high-resolution images for mapping and surveying.
- **Environmental Monitoring:** Monitoring environmental changes and natural disasters using image analysis.

## 3. Industrial Inspection

- **Quality Control:** Automating the inspection process to ensure product quality and consistency.
- **Defect Detection:** Detecting defects in manufacturing processes to maintain high standards.

- **Robotics Vision:** Enabling robots to interpret and navigate their environment using image processing techniques.

## 4. Security and Surveillance

- **Facial Recognition:** Identifying individuals by analyzing facial features for security purposes.
- **Object Detection:** Detecting and identifying objects in surveillance footage to enhance security measures.
- **Motion Detection:** Monitoring and detecting movement in video feeds for security and surveillance.

## 5. Automotive Industry

- **Autonomous Vehicles:** Processing images from sensors to enable autonomous driving.
- **Traffic Sign Recognition:** Identifying and interpreting traffic signs to assist drivers and autonomous systems.
- **Driver Assistance Systems:** Enhancing driver safety with features like lane departure warnings and collision avoidance.

## 6. Entertainment and Multimedia

- **Photo and Video Editing:** Enhancing and manipulating images and videos for artistic and practical purposes.
- **Virtual Reality and Augmented Reality:** Creating immersive experiences by integrating real-world images with virtual elements.
- **Gaming:** Enhancing graphics and creating realistic environments in video games.

## 7. Document Processing

- **OCR (Optical Character Recognition):** Converting printed text into digital text for easy editing and searching.
- **Barcode and QR Code Scanning:** Reading and interpreting barcodes and QR codes for quick information retrieval.
- **Document Enhancement and Restoration:** Improving the quality of scanned documents and restoring old or damaged documents.